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Safeguarding cyberspace: Enhancing malicious website detection with PSO—optimized XGBoost and firefly-based feature selection

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ABSTRACT

In recent years, the exponential growth of internet usage worldwide has created a conducive environment for the expansion of malicious activities. Among these threats, malicious websites pose a significant risk to individual users and corporations. This paper presents a robust and efficient model for the detection of various types of malicious websites with high accuracy in the process. The proposed approach employs a two-step process. Firstly, a feature selection method based on the Firefly algorithm is utilized to identify the most relevant features. Subsequently, an optimized version of the XGBoost algorithm is applied to classify websites based on the selected features. Optimization of XGBoost's parameters is achieved through the Particle Swarm Optimization (PSO) algorithm to enhance its performance. To assess the efficacy of the introduced model, the model is evaluated against several benchmark classification algorithms using a dataset comprising over 36,000 websites. In binary classification, the introduced model surpasses other benchmark methods with a significant 98.42 % classification accuracy and an F1 score of 0.984. For multiclass problem classification, it consistently achieves over 98 % accuracy in each class. The test results highlight the proposed model's robust performance, characterized by exceptional classification accuracy and F1-measure rates. It demonstrates the capability to detect various types of malicious websites with high precision and minimal false error rates.

1. Introduction

The rapid evolution of the internet has significantly boosted the number of businesses transitioning to an online presence (Grubor and Jakša, 2018). This rapid growth in online business has also led to an alarming surge in cybercrimes perpetrated through malicious websites (Kakol et al., 2017). Consequently, detecting and analyzing these disreputable web pages have emerged as critical challenges within the online environment. Even expert individuals who are mindful of malicious activities have fallen victim to such deceptive websites.

Typically, malicious web pages are used in online attacks aimed at pilfering users' sensitive information. Perpetrators create fraudulent websites that mimic legitimate ones, leading users to believe they are interacting with a genuine site when, in fact, they are navigating a malicious one (Basit et al., 2021). Currently, a variety of malicious websites exist, disguising themselves as anything from phishing sites to platforms for hosting or disseminating malware (Wang et al., 2023).

Among these malicious activities, phishing websites stand out as a particularly prevalent threat on the internet (Jeeva and Rajsingh, 2016). Phishing's primary objective is to gain access to sensitive information by crafting counterfeit web pages that closely resemble legitimate ones, such as those of banks or renowned corporations (Salahdine and Kaabouch, 2019). Once users access these fake sites, they are prompted to enter sensitive data, such as social security numbers (SSNs), credit card information, and other personal credentials (Harinahalli Lokesh and BoreGowda, 2021). Malicious poses a severe threat to ordinary users and undermines the trust and reputation of commercial enterprises, as users become increasingly apprehensive about using online services without falling prey to phishing schemes (Jeeva and Rajsingh, 2016). Detecting malicious websites plays a pivotal role in safeguarding internet users.

Fraudsters employ a range of tactics, including technical and social engineering methods, to illicitly obtain users' sensitive information (Zhang et al., 2017). While numerous methods have been introduced in recent years to mitigate online malicious attacks, these threats are expected to become more sophisticated in the near future (Dixit et al., 2021). Consequently, addressing the malicious website problem necessitates a continually evolving solution capable of identifying modern malicious websites. Such solutions should contain various measures,

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including technical, legal, and educational strategies; however, no single comprehensive method can detect all forms of malicious URLs (Adebowale et al., 2019).

Security companies have invested considerable effort and resources in shielding users from malicious sites. Despite ongoing advancements in security methods, criminals continue to employ increasingly sophisticated techniques (Sheikhi, 2020). Online protection methods are designed to preemptively shield users from interacting with malicious websites, achieved through precise identification of such web pages with minimal processing time and the implementation of effective warning systems (Ramesh et al., 2017).

In recent years, the ease of creating malicious websites has escalated, underscoring the urgent need for a robust online protection system (Sheikhi, 2021). Consequently, many researchers have turned to machine learning techniques to combat the issue of malicious websites and enhance web page detection. However, most existing methods have focused on individual web pages, necessitating feature extraction. These techniques rely on the attributes of single web pages, including images and text features, rendering them ill-suited for comprehensive investigations across entire websites.

In response to the critical need for effective detection of malicious URLs, this study introduces a cutting-edge model that stands out by innovatively distinguishing between features of malicious and benign websites. Specifically, the proposed method employs the firefly optimization algorithm to enhance feature selection, eliminating irrelevant and redundant attributes. Subsequently, the method employs Extreme Gradient Boosting (XGBoost) to classify malicious websites within the refined feature set. The proposed PSO-XGBoost model combines the XGBoost algorithm with swarm intelligence optimization. Within this model, we leverage the exceptional search capabilities of the Particle Swarm Optimization (PSO) algorithm to fine-tune the hyperparameters of the XGBoost model. This strategic optimization, specialized for the complexities of web security threats, is balanced to set a new benchmark in classification accuracy and model adaptability. However, The main contributions of our study are summarized as follows:

- Development of a PSO-XGBoost model, leveraging the XGBoost algorithm and Particle Swarm Optimization (PSO) for adaptive parameter optimization, aiming to detect various malicious websites.
- Implementation of a feature selection technique based on the Firefly algorithm strategically chosen to curate an optimal feature set. This enhances classification accuracy while reducing processing time.
- A ranking of features key for the identification of malicious web pages, thereby providing insights into their relative importance within the XGBoost model, and contributing to a more interpretable machine learning process.
- A comprehensive evaluation of the proposed model's performance using a dataset containing more than 36,000 websites. Comparative analysis conducted against state-of-the-art classification algorithms.
- Demonstration of the superior efficacy of the proposed model over benchmark classification algorithms, achieving a remarkable accuracy rate of over 98% in website classification.

The rest of the paper is structured as follows: Section 2 provides a comprehensive literature review and examines the outcomes of various methods proposed in recent years for detecting malicious websites. Section 3 serves as the methodology section, elucidating details regarding the dataset employed alongside a thorough description of the feature selection approach and the application of the Particle Swarm Optimization (PSO) algorithm. Section 4 introduces the PSO-XGBoost model, presenting its conceptual foundation and operational intricacies. Section 5 is dedicated to expounding the experimental setup, along with a comprehensive analysis of the results yielded by the proposed model. Finally, in Section 6, we bring the paper to a close, encapsulating key findings and providing a conclusive summary of the research endeavor.

2. Related works

The internet serves as a conducive environment for malicious online activities, with malicious actors leveraging it to propagate harmful websites aimed at gathering sensitive data, including bank credentials, passwords, credit card details, and private information. The presence of such deceptive web pages poses a significant threat to both individual users and companies operating in the online domain (Han et al., 2019). Consequently, many researchers have dedicated their efforts to exploring diverse solutions for detecting these malicious web pages. This section reviews and analyzes select solutions developed in recent years to address this pressing issue.

Ali and Ahmed (2019) proposed a new method based on classification algorithms for detecting malicious websites using the genetic algorithm (GA) and deep neural networks (DNNs). In the proposed hybrid approach, the most significant subset of features and an optimal set of the weight of websites feature were discovered based on the genetic algorithm (GA) to increase the model's accuracy. Then, the identified features have been used to train DNNs to predict phishing websites accurately. The experimental results show that the proposed model obtained higher classification accuracy and a lower error rate than the other algorithms. In another work, Wang et al. (2019) implemented a new method for detecting malicious webpages using a Convolutional Neural Network (CNN) and an independent recurrent neural network called CBIR. The authors extracted and merged some website features, such as the URL word vector feature, "texture fingerprint" feature, and URL static vocabulary. Then the merged features were analyzed and used to identify malicious web pages based on the proposed CBIR method. The results of the comparison study with other techniques show that the proposed model obtained good performance in terms of accuracy and improved the performance of the malicious webpages model. Shin et al. (2022) proposed an ensemble ML strategy to improve the detection of malicious webpages, surpassing the limitations of single models reliant on specific features. Their approach involved advanced feature definition, training and testing with six ML models, and combining predictions using a weighted soft voting ensemble. Results demonstrated a 6 % performance increase, detecting 141 more malicious URLs. This ensemble framework, incorporating advanced features, effectively anticipates evolving threats, outperforming single models.

Ibrahim and Hadi (2017, October) have introduced a method for phishing website prediction. At first, they eliminated irrelevant features from the dataset, then employed several different classifiers to study their capabilities in the classification of phishing websites. Finally, in their experiments, the introduced methodology achieved good performance in terms of accuracy. In another study, Akiyama et al. (2017) proposed a monitoring system based on a honeypot technology. The proposed system suggests useful countermeasures for malicious URL redirection based on the information achieved by monitoring URL redirection behavior. Then, the security operators take achieved information by the honeypot-based monitoring system to keep the network safe. Abdelhamid et al. (2014) developed a novel model to detect phishing websites using associative classification. The authors have used the MCAC method in their approach in order to identify features that help to recognize phishing websites. The proposed method outcomes determined that the MCAC algorithm performed more promising than other data mining techniques regarding classification accuracy.

Peng et al. (2019) presented a new method to detect malicious URLs, which is called Attention-Based CNN-LSTM and is based on the attention mechanism with CNN and Long Short-Term Memory Network (LSTM). The authors used the whois check service to extract and filter features such as URLs, Whois information, string statistical information, and texture information. Then, the features are pre-processed and utilized as input of CNN and LSTM algorithms to calculate local and global features. Finally, the method classified URLs based on global features and the SoftMax function. The observed results show that the Attention-based CNN-LSTM mechanism detected malicious URLs more accurately than

other techniques. in another research, Hassan et al. (2022) developed a machine-learning approach to identify malicious websites, enhancing digital security. Their method used undersampling, oversampling, and SMOTE techniques to balance the data and reduce bias. Evaluation of various ML algorithms revealed that the random forest algorithm, with oversampling, produced the most effective results. This study highlights the critical role of data balance and quality in improving website classification for strengthened digital security. Ubing et al. (2019) presented a method for phishing website detection using the ensemble technique. The research proposes an accurate phishing website detection method using feature selection and a voting ensemble algorithm. Firstly, they selected high-ranked features on the dataset. They have then employed a majority-based voting technique to identify phishing websites on a dataset. The experiment results show that the introduced method accurately predicted the class of almost 95 % of websites.

Awotunde et al. (2021) proposed an Anomaly Detection System (ADS) model for detecting malicious activities in IIOT network traffic using Deep learning techniques called DAE-DFFNN. Their system employed a hybrid rule-based method to reduce the dimensionality of data captured from TCP/IP packets. Then, it applies the deep feedforward neural network model to detect destructive activities. They validated the proposed model on two well-known datasets named NSL-KDD and NSW-NB15. The test results demonstrate that the method has good performance in identifying the malicious activities in the network with a low false alarm rate. Sahingoz et al. (2019) introduced a real-time anti-phishing system to solve phishing URLs problems based on natural language processing (NLP) based features and several classification algorithms. The authors constructed a new dataset for validation of the proposed system. The experimental results show that among the implemented algorithms, the random forest algorithm was more successful in detecting phishing URLs using NLP-based features, distinguishing over 97 % of URLs accurately. Jabri and Ibrahim (2015) proposed an improved version of the PRISM Algorithm to identify phishing web pages. They improved the PRISM to generate all absolute rules at the same time. The experiment results showed their modified method outperformed the original PRISM algorithm in terms of classification accuracy, a lower error rate, and the number of generated rules. Mamun et al. (2016) introduced a system to detect malicious websites by using lexical analysis. They have provided a valuable dataset of different types of malicious attacks with 79 features. We also used this dataset in our work. They showed that lexical analysis is helpful in the detection of malicious URLs.

A primary challenge associated with existing methods is the handling of high-dimensional training data, which substantially increases processing time and utilizes significant system resources Mamun et al. (2016, September). Moreover, the selection of an appropriate classification technique plays a critical role in enhancing the detection rate of unknown instances. These issues have prompted our research to introduce a novel model for the automatic detection of malicious websites, thereby addressing previous limitations. In this study, we adopt a firefly-based feature selection approach to identify the most optimal feature set, combined with the application of the proposed PSO-XGBoost model. This model demonstrates outstanding performance in accurately identifying various types of malicious URLs.

3. Methodology

The proposed method aims to build a novel model that can effectively filter malicious websites with a high detection rate. This section describes implementation details, dataset, features, and the summary of the proposed model.

3.1. Data collection

In this research, we evaluated the proposed model using a malicious URL dataset sourced from URL dataset (2024). The dataset employed in our experiments comprises a total of 36,707 malicious URLs (Mamun et al., 2016, September), accompanied by 79 distinct numerical features designed for the identification of malicious websites. These features exhibit varying ranges of values, underscoring the diversity inherent in the dataset.

Furthermore, the malicious URLs within the dataset have been categorized into four principal classes based on their respective attack types, including spam, phishing, malware, and defacement URLs. A comprehensive summary of the dataset, including its dimensions and class distribution, is provided in Table 1.

In the employed dataset, five different types of URLs are considered, the process of collecting each of these five different categories explained as follows:

Benign URLs: all of the benign URLs in the dataset were obtained from Alexa top websites.

Spam URLs: the spam URLs in the dataset were taken from the WEBSPAM-UK2007 dataset, which is available publicly on the internet.

Phishing URLs: the phishing URLs in the dataset were collected from a repository of active phishing webpages, which is called OpenPhish.

Malware URLs: The list of malware sites on the dataset was taken from DNS-BH, which is a project that creates and keeps a list of malware and spyware webpages.

Defacement URLs: approximately 21 percent of URLs in the dataset belong to the defacement URL category. All of these URLs were taken from hidden URL that includes malicious websites or trusted hosting fraudulent websites which are ranked on Alexa.

3.2. Pre-processing phase

The pre-processing phase assumes a crucial role in our proposed model. During this phase, we executed several operations on the dataset with the aim of enhancing its quality and, consequently, obtaining more reliable results in both the training and testing phases. These processes contained the following steps:

- **Rescaling and Normalization:** The features within the dataset span a wide range of values, causing them to be incomparable. To address this issue and ensure better performance, we standardized the dataset by rescaling all feature values to a uniform [0–1] range. This operation is performed once, following the implementation of the model, and is known to have a positive impact on the training and predictive accuracy of machine learning algorithms (Sheikh Khozani et al., 2020)
- **Handling Missing Values:** In practice, many datasets contain missing values within their features, stemming from a variety of causes. While missing values may not necessarily impair predictive models in all cases, they tend to adversely affect the quality of results in the majority of models, posing a threat to the reliability of predictions. In order to enhance the performance and trustworthiness of our model, we addressed this issue by substituting missing values with the mean or mode values of each feature.

Table 1
Description of the dataset.

	Records	Percentage
benign	7781	21.60
Defacement	7930	21.20
Malware	6712	18.29
Phishing	7586	20.67
Spam	6698	18.25
Total	36,707	100

3.3. Feature selection

Feature selection is an essential phase in the development of a predictive model, especially in domains where precision is as critical as in the identification of malicious websites. An optimal feature set not only sharpens the model's accuracy but also enhances computational efficiency by eliminating irrelevant or redundant attributes that could otherwise lead to overfitting and increased processing time (Sheikhi, 2020).

To effectively address these challenges within our dataset, we adopted a hybrid feature selection strategy that combines the CFSSubsetEval method with the Firefly algorithm, with the goal of isolating the most impactful subset of features for the accurate categorization of websites.

The CFSSubsetEval method initiates this strategy by evaluating subsets of features based on their predictive ability and inter-feature redundancy. This process ensures that we retain features with substantial individual predictive power while discarding those that are redundant, thus enhancing the model's interpretability and generalization capabilities.

Subsequently, the Firefly algorithm, a meta-heuristic inspired by the natural flashing patterns of fireflies, optimizes this curated subset by employing its capability to traverse complex search spaces efficiently. This algorithm is particularly adept at locating not just local optima but also at seeking global solutions, an essential feature for circumventing the pitfalls of local minima, which can be a significant concern in feature selection tasks.

Through an extensive analysis employing this hybrid approach on the initial set of 79 features, we identified 23 features that demonstrated a noticeable correlation with the classification labels, suggesting their significant role in the model's predictive accuracy. The following criteria guided the selection of these features:

- **Predictive Power:** We precisely tested each feature for its contribution to accurate predictions. Only those features that consistently enhanced the model's predictive performance were selected.
- **Uniqueness:** The selected features were instructed to provide distinct and non-overlapping information. This criterion ensured the elimination of redundant features and the conservation of those that offered unique insights into the dataset.
- **Computational Efficiency:** In order to enhance precision, we also preserved an emphasis on computational efficiency. We favored and considered features that strengthened the model's efficiency, speeding up training and prediction phases without sacrificing accuracy.

These criteria were influential in clarifying a feature set that is not only statistically valid but also highly relevant and efficient for identifying malicious websites. The combination of CFSSubsetEval and the Firefly algorithm produced validated and practically assessed selection, ensuring that our model operates with boosted accuracy and efficiency.

Table 2 presents the specific features that were selected after a strict selection process, along with the justification for their inclusion. This table provides a transparent view of the relevance of each feature, highlighting their individual and collective contributions to the model's performance in malicious website detection.

Table 2
List of selected features and their rationale for malicious website detection.

No	Feature name	Rationale/Relevance
1	Querylength	Longer queries in URLs may indicate obfuscation attempts, common in malicious sites.
2	domain_token_count	Higher count may suggest domain complexity, often seen in phishing attempts.
3	avgpathtokenlen	Unusually long or short path tokens can be indicative of a malicious URL structure.
4	tld	Certain top-level domains are more frequently used in malicious websites.
5	charcompvowels	A high proportion of vowels could relate to domain generation algorithms.
6	urlLen	Malicious URLs often use longer lengths to hide suspicious parts.
7	domainlength	Similar to urlLen, longer domains may harbor malicious content.
8	fileNameLen	Executable file names with long lengths can be associated with malware.
9	argDomanRatio	A high ratio may indicate a suspicious number of arguments, often seen in attacks.
10	executable	Presence of executable files could be a direct indicator of malicious intent.
11	NumberofDotsinURL	Excessive dots may be used for subdomain trickery, common in phishing sites.
12	CharacterContinuityRate	High rates may indicate domain generation algorithms or attempt to mimic brands.
13	LongestVariableValue	Longer values may suggest encoded or obfuscated malicious content.
14	Arguments_LongestWordLength	Long argument words can be a sign of code injection or complex query parameters.
15	spcharUrl	Special characters are often used in URL to obfuscate malicious links.
16	delimeter_Domain	Use of unusual delimiters in domain names could signify suspicious URL structuring.
17	delimeter_path	Similar to delimeter_Domain, could indicate abnormal path structures.
18	NumberRate_Domain	Numerical domains are less common and can be used for deceptive purposes.
19	NumberRate_DirectoryName	High numerical rates in directory names may signal an attempt to hide payloads.
20	NumberRate_AfterPath	Numerical information post-path can be associated with tracking or malicious redirects.
21	SymbolCount_Domain	Excessive symbol use in domains can be a tactic to confuse users or hide real addresses.
22	SymbolCount_Extension	Uncommon symbol usage in extensions may be related to malicious file types.
23	Entropy Domain	High entropy indicates randomness, often a characteristic of malicious domains.

3.4. Extreme gradient boosting

The Extreme Gradient Boosting (XGBoost) algorithm has been used in the proposed classification model to identify malicious websites with the highest accuracy. This XGBoost algorithm is a scalable implementation of the gradient boosting framework, which was initially proposed by Friedman (2001 and 2002) and is one of the most efficient algorithms for solving supervised learning tasks such as classifications and regression (Friedman, 2001; Friedman, 2002). The XGBoost uses precise control of the variable weights and is referred to as a regularized gradient boosting technique. This algorithm improves classification performance by reducing overfitting by utilizing a more regularized model formalization (Carmona et al., 2019).

In recent years, the XGBoost attained more attraction and appeared to be influential for many different purposes. Many data scientists use it in order to produce state-of-the-art results for different machine learning challenges. One of the significant reasons for XGBoost success is its scalability in different challenges and scenarios (Chen and Guestrin, 2016, August). The XGBoost system runs significantly faster than current solutions, and it scales to billions of samples on a device. Therefore, as signified by the name, the core concept of the XGBoost algorithm is boosting. This algorithm can transform weak classifiers into robust ones using additive training strategies and integration (Adnane et al., 2019, March).

The boosting reduces overall errors by introducing supplementary models based on errors in previous iterations, which improves its classification performance and makes XGBoost one of the most efficient

algorithms for classification problems (Carmona et al., 2019). In general, XGBoost is the base of the tree and linear base learners. In this study, we used a linear base XGBoost algorithm called Xgblinear.

3.5. Particle swarm optimization (PSO) algorithm

Particle swarm optimization (PSO) is an optimization algorithm introduced by Eberhart, Kennedy. This algorithm is inspired by the research of social animals like fish or birds swarms foraging behavior. It is also classified as a metaheuristic algorithm. The main idea of it is to produce more useful social information among individuals in a crowd. In PSO, particles are social animals like birds or fishes, and every particle can search in the N-dimensional search space.

In a searching procedure, each particle holds two properties named position and velocity. Position indicates the direction of an individual particle, and velocity denotes the movement speed of particles. In general, the current position of an individual particle in the crowd can be the solution. Generally, the best solution discovered by each particle is regarded as the individual optimal, and the best solution among all particles is taken as the global optimal. During the search process, the position and velocity attributes are continuously updated, and particles exchange experiences and information to update better positions. Finally, when the algorithm reaches the termination conditions, it exits the iteration. The PSO algorithm implements for optimal searching can be described as follows:

Randomly initialize the population, velocity of particles, and position of particles in search space.
Define fitness function. After that, each particle search for an optimal solution in search space with the initial velocity received in the first step. Global optimal also produced form optimal solution of each particle. Then the current global optimal solution determines to update after comparison with historical global optimal.
The velocity and position of each particle are updated as shown as follows:

$$V_{id} = \omega V_{id} + C_1 Random(0,1)(P_{id} - X_{id}) + C_1 Random(0,1)\left(P_{gd} - X_{id}\right) \quad (1)$$

$$X_{id} = X_{id} + V_{id} \quad (2)$$

- Where *C1* and *C2* indicate social learning and individual factors of particles, *P* id denotes the *d-th* dimension of the local best of the *i th* particle, while P_{gd} represents the *d-th* dimension of global best. ω expresses coefficient of inertial weight, and its linearly reducing procedure can be described as

$$\omega = \frac{\omega_{max} + (iter - iter_i) \times (\omega_{max} - \omega_{min})}{iter} \quad (3)$$

- Where *iter* represents the maximum number of iterations while $iter_i$ indicates the current number of iterations and ω min, and ω_{max} in equation expresses the minimum and maximum values of ω.

4. PSO-XGBoost model

This section introduces the PSO-XGBoost model, which represents a novel approach that combines the strengths of the PSO algorithm and eXtreme Gradient Boosting (XGBoost) for predicting malicious websites. The hybrid model integrates the powerful classification capabilities of the XGBoost algorithm with the global optimization efficiency of PSO. Specifically, we employ the XGBoost's linear booster (XgbLinear) for its ability to manage sparse data and its computational efficiency, which is particularly important when dealing with large datasets standard in web security applications.

4.1. Model innovation

The innovative application of PSO in this context lies in its capability to conduct a global search for the optimal set of hyperparameters of the XGBoost model, which is not typically seen in standard applications. Unlike grid search or random search methods, PSO navigates the parameter space by simulating a social behavior pattern akin to a flock of birds or a school of fish. This social heuristic is particularly powerful in avoiding local optima and efficiently finding the best solution in a complex, multidimensional hyperparameter landscape.

Furthermore, the application of PSO to the XgbLinear method is less common compared to the XgbTree. The XgbLinear method benefits significantly from parameter optimization, especially when dealing with sparse data or features with linear relationships. The optimization targets multiple hyperparameters, such as the number of boosting iterations, learning rate, and regularization terms, which are critical to the model's performance. Our model innovatively adjusts these parameters harmoniously to achieve superior predictive performance.

4.2. Hyperparameter optimization

The XgbLinear model contains a range of parameters, which were systematically optimized in this study using the PSO algorithm, aimed at enhancing the classification performance of the model.The performance of the XgbLinear model depends heavily on its hyperparameters, which include the number of boosting iterations (nrounds), learning rate (eta), and regularization terms (lambda for L2 and alpha for L1). To ensure optimal performance, these parameters are meticulously optimized using the PSO algorithm. The detailed information about these parameters is presented in Table 3.

This table presents the default values, permissible ranges, and function of each hyperparameter in the XgbLinear model. The optimization process aims to maximize the detection rate of malicious websites by determining the most effective combination of these parameters.

4.3. Model development

Fig. 1 illustrates the development scheme of the PSO-XGBoost model. The process begins with the collection of website URLs, followed by feature selection based on the data characteristics and the threat landscape. The dataset is then split into training (80 %) and testing (20 %) sets to ensure the model is evaluated on unseen data. The structure of the development of the PSO-XGBoost model for malicious website detection is shown in Fig. 1.

As shown in Fig. 1, the proposed PSO-XGBoost model development contains the following steps:

1. **Dataset Splitting:** Dividing the collected data into training and testing datasets.
2. **Feature Selection and Initialization:** Selecting features and initializing the parameters of the XgbLinear model.
3. **PSO Algorithm:** Launching the PSO for hyperparameter optimization, including the calculation of particle fitness and updating their velocity and position to find the local and global bests.

Table 3
Parameters of the XgbLinear Model Optimized Using PSO.

Parameters	Default value	Range	Explanation
nrounds	100	[10, ∞]	number of boosting iterations
eta	0.3	[0,1]	learning rate
lambda	0	(0,1]	It enables Ridge Regression
alpha	1	(0,1]	It enables Lasso Regression

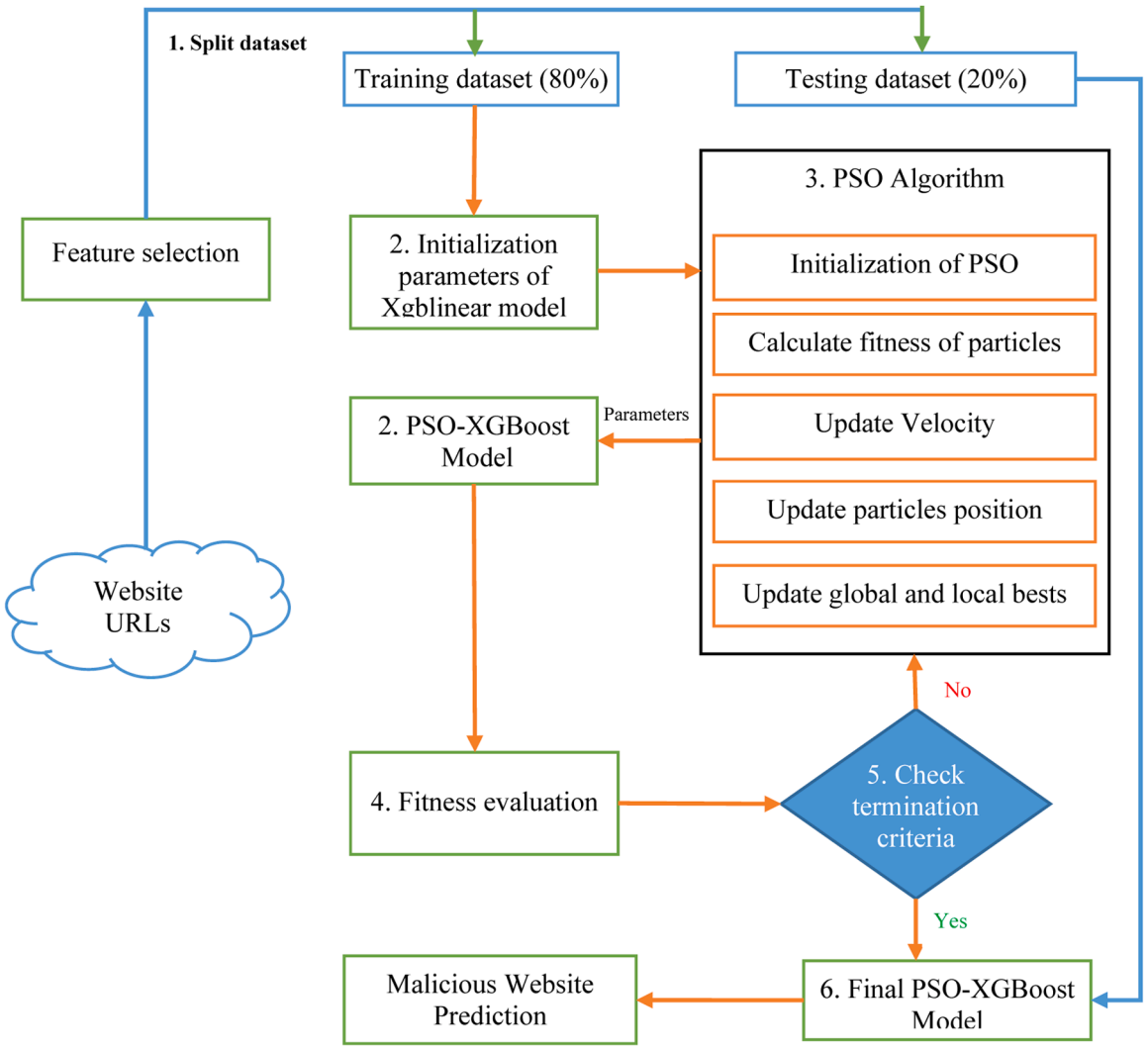


Fig. 1. Scheme of the development of the PSO and PSO-XGBoost model.

4. **Fitness Evaluation:** Assessing the model's performance iteratively to adjust the parameters effectively.
5. **Termination Check:** Determining if the optimization process has met the termination criteria, such as maximum iterations or desired accuracy.
6. **Final Model Deployment:** Applying the optimally configured PSO-XGBoost model to the testing dataset for malicious website prediction.

The structured optimization process is designed to significantly enhance the model's classification capability. Unlike traditional grid or random search methods, PSO's metaheuristic search provides a robust mechanism for navigating through the hyperparameter landscape. The XGBoost model is initialized with its default parameters, and then the PSO algorithm takes over to optimize these parameters. Particles are used to represent potential solutions, and they navigate through the hyperparameter space, adjusting their positions based on their individual experiences and the collective knowledge of the swarm. The objective is to maximize the fitness function defined by the model's accuracy in predicting malicious websites.

4.4. Fitness function and algorithm convergence

The fitness of each particle is evaluated using a fitness function that is a composite measure including accuracy, precision, recall, and the F1-score. The PSO algorithm iteratively updates the velocities and positions of the particles until the termination criteria, such as a maximum number of iterations or a target error threshold, are satisfied.

4.5. Malicious website prediction

After meeting the termination criteria, the final PSO-XGBoost model is finalized. This model is then used to evaluate and predict the maliciousness of websites. The performance of the model in classifying malicious websites is verified using the testing dataset. This process, as illustrated in Fig. 1, ensures a comprehensive search for the most effective model configuration. It highlights the adaptability and strength of the PSO algorithm in fine-tuning the parameters of the XgbLinear model. This leads to an improvement in the model's ability to accurately identify malicious web entities compared to a model with non-optimized parameters.

5. Experiments and result analysis

This section contains the experimental framework, the examination of experimental outcomes of the proposed model. Subsection 5.1 elucidates the experiment's preparation and methodology. Subsection 5.2 delineates the performance evaluation metrics employed to assess the proposed method's effectiveness. Lastly, Subsection 5.3 presents an analysis of the results obtained by the model to measure its efficacy in

detecting various categories of malicious websites. Additionally, the achieved results are subjected to a comparative analysis against outcomes obtained from alternative techniques.

5.1. Experiment setup

The implementation of the PSO-XGBoost model was carried out on a system equipped with 16 GB of RAM, utilizing the R programming language and the Caret package. Within our model, we initially used feature selection to identify the optimal subset of attributes. Subsequently, the dataset was partitioned into separate training and testing sets. Specifically, 80 % of the data was allocated to the training set, while the remaining 20 % was reserved for evaluating the model's performance. To enhance model robustness and reduce potential errors, a 10-fold cross-validation methodology was employed during the training phase.

Additionally, the PSO algorithm's key parameters encompass the number of particles, inertia weight, maximum iterations, global and local learning factors, and the dimensions of each particle. A comprehensive listing of these parameter settings is provided in Table 4.

5.2. Evaluation metrics

In the experiments, for better evaluation effectiveness of the proposed model, we have used several well-known metrics, namely, the true positive rate (TPR), the false positive rate (FPR), F-measure, precision, recall, and classification accuracy (ACC). The value of the TPR measured by the Eq. (4)

$$TPR = \frac{TP}{P} = \frac{TP}{(TP + FN)} \tag{4}$$

In the Eq. (4) P is the representative of the total number of malicious pages, and the TP is representative of records which are correctly classified as malicious sites. P also is equal to the value of TP and the number of records incorrectly identified as malicious websites (FN). The FPR also representative estimates the percentage of records which incorrectly identified as malicious sites. The value of FPR is measured by Eq. (5).

$$FPR = \frac{FP}{L} = \frac{FP}{(FP + TN)} \tag{5}$$

In the Eq. (5) L is the representative of the total number of legitimate web pages, and the FP is representative of records which are incorrectly classified as malicious sites. Moreover, L also is equal to the value of FP and the number of records correctly identified as legitimate websites (TN). The classification accuracy (ACC) estimates the closeness among evaluations of forecasted records and the total number of legitimate and malicious websites. The value of ACC is measured by Eq. (6).

$$ACC = \frac{(TP + TN)}{(P + L)} \tag{6}$$

False Negative Rate (FN): It represents the percentage of malicious websites that were incorrectly classified as legitimate by the proposed model. The S denotes the total number of malicious websites while T represents the number of malicious websites that were incorrectly classified as legitimate, the value of FN is measured using Eq. (7).

$$FN = \frac{T}{S} \tag{7}$$

Precision: It represents the percentage of websites that were classified correctly as malicious by the proposed model. It presents the accuracy of the proposed model. The value of precision is measured using Eq. (8).

$$Precision = \frac{TP}{TP + FP} \tag{8}$$

Recall: It describes the percentage of malicious websites that were correctly classified as malicious. It presents the completeness. It calculated using Eq. (9).

$$Recall = \frac{TP}{TP + FN} \tag{9}$$

F-measure: It is defined as the harmonic mean of precision and recall. It calculated using Eq. (10).

$$F - measure = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{10}$$

5.3. Result analysis

To comprehensively evaluate the performance of our proposed model, we conducted a series of experiments and compared the results with those obtained using several established classification algorithms. Our proposed model and the benchmark techniques were implemented under identical system configurations and subjected to testing on the same dataset, ensuring a fair and equitable comparison. The optimal model configuration was determined through the optimization process, yielding the following settings: eta (learning rate) of 0.3861141, alpha (L1 regularization term) of 0.01339033, lambda (L2 regularization term) of 0.382388, and nrounds (number of boosting rounds) set to 275. Subsequently, the PSO-optimized XGBoost model was applied to the testing dataset.

Table 5 provides a detailed presentation of the achieved results for our proposed model and other benchmark classification algorithms. The PSO-XGBoost model demonstrates an impressive classification accuracy of 98.42 %, a weighted average precision rate of 0.984, a Kappa statistic of 98.02 %, and a weighted average F-measure rate of 0.984.

The primary aim of our conducted experiments is to validate the performance of the proposed model in the detection of malicious websites. The results distinctly indicate the superiority of our approach in binary classification, as evidenced by its superior classification accuracy and F-measure rates compared to other classification algorithms. Following our model, the J48 and Random Tree algorithms achieved considerable accuracies of 94.91 % and 94.61 %, respectively. However, a substantial performance gap remains between these algorithms and our proposed model. In contrast, the Naive Bayes algorithm produced the least favorable results, mainly due to its inherent assumption of feature independence.

Furthermore, these findings underscore the satisfactory performance of the selected features in our proposed model, highlighting the significant role played by the feature selection method in enhancing the

Table 4
Parameter settings of PSO.

Parameters	Value
inertia weight	(0.4,0.9)
max iterations	100
global learning factor C1	1.193
local learning factor C2	1.193
particles dimension	990
Particles number	10

Table 5
Comparisons between the proposed method and several classification techniques.

Algorithm	Accuracy	Kappa	Avg Precision	Avg F1	Avg Recall
J48	94.91	93.64	0.949	0.949	0.949
Random Tree	94.64	93.30	0.946	0.946	0.946
Naive Bayes	56.42	45.14	0.584	0.564	0.542
MLP	80.64	75.75	0.807	0.806	0.806
RBF	60.64	50.52	0.613	0.606	0.594
PSO-XGBoost	98.42	98.02	0.984	0.984	0.984

overall classification performance. Subsequently, we conducted a comprehensive comparative analysis, detailed in Table 6, to evaluate the results of our proposed model against those of other classification algorithms in the context of multiclass classification.

The primary objective of the multiclass classification test is to assess the model's efficacy in identifying malicious websites across various classes within the dataset. As depicted in Table 6, the PSO-XGBoost model demonstrated the highest success rates across all classes, achieving impressive percentages of 99.43, 99.71, 99.43, 98.54, and 99.73 for the respective classes. Following the proposed model, both J48 and Random Tree also exhibited robust performance in all classes, with minimal deviation from the PSO-XGBoost model's results. Notably, all classification algorithms, except for Naive Bayes, displayed superior performance in classifying websites under the "spam" category compared to other classes.

Moreover, a comprehensive analysis of the results indicates that tree-based algorithms outperformed network models, producing the highest average accuracy across all classes. While the RBF and Naive Bayes models demonstrated comparable classification accuracy in most classes, the findings presented in Table 6 highlight the superior performance of the MLP model over the RBF model, particularly concerning the F-measure rate.

The Confusion Matrix (CM) comprehensively represents the model's performance by detailing the correspondence between actual and predicted classifications. In multiclass classification, the CM serves as a valuable tool for assessing whether the model effectively identifies all classes or overlooks specific categories. Fig. 2 displays the confusion matrix for the proposed model. The presented confusion matrix furnishes crucial information concerning overall accuracy, True Positive Rate (TPR), and False Positive Rate (FPR) for all classes within the dataset.

The presented confusion matrix demonstrates that the suggested model correctly classifies websites in the "Defacement" class, while its performance is relatively weaker in the "Malware" category. These results underscore the model's capacity for high classification accuracy

Table 6
Comparison performance of the model and different algorithms in the detection of different type of malicious websites.

Algorithm	Class	Accuracy	Precision	F1	Recall
J48	benign	98.57	0.97	0.96	0.97
	Defacement	98.39	0.98	0.95	0.96
	malware	98.34	0.95	0.96	0.95
	phishing	95.94	0.88	0.92	0.90
	spam	98.60	0.96	0.96	0.96
Random Tree	benign	98.41	0.96	0.96	0.96
	Defacement	98.36	0.97	0.95	0.96
	malware	98.08	0.96	0.94	0.95
	phishing	95.57	0.88	0.91	0.89
	spam	98.98	0.97	0.98	0.97
Naive Bayes	benign	85.33	0.85	0.60	0.71
	Defacement	81.07	0.64	0.56	0.59
	malware	80.77	0.38	0.47	0.42
	phishing	80.81	0.65	0.54	0.59
	spam	84.88	0.24	0.77	0.37
MLP	benign	92.02	0.93	0.75	0.83
	Defacement	92.30	0.93	0.75	0.83
	malware	91.66	0.69	0.83	0.75
	phishing	88.87	0.79	0.71	0.75
	spam	96.43	0.90	0.90	0.90
RBF	benign	85.53	0.88	0.60	0.72
	Defacement	83.42	0.64	0.61	0.63
	malware	84.53	0.35	0.64	0.45
	phishing	81.81	0.63	0.56	0.59
	spam	86.00	0.48	0.65	0.56
PSO-XGBoost	benign	99.43	0.99	0.99	0.99
	Defacement	99.71	0.99	0.99	0.99
	malware	99.43	0.99	0.98	0.98
	phishing	98.54	0.96	0.97	0.96
	spam	99.73	1.0	0.99	0.99

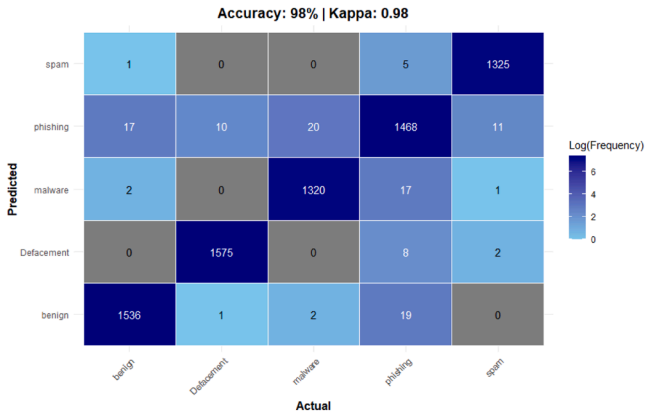


Fig. 2. Confusion matrix of proposed model.

across all classes, complemented by a notably low false-positive rate.

The Receiver Operating Characteristic (ROC) curve serves as a vital evaluation metric for visually assessing a model's performance and determining its suitability in terms of cost sensitivity. Additionally, the Area Under the Curve (AUC) measures the model's ability to distinguish between classes. Fig. 3 displays the AUC-ROC curve for each class.

The ROC curve is constructed by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR), with TPR on the y-axis and FPR on the x-axis. Fig. 3 offers insights into the model's ability to differentiate between various classes and quantifies its separability. Higher AUC values signify the model's effectiveness in class discrimination. An AUC value close to 1 indicates strong separability, while proximity to 0 suggests limited separability. The data in Fig. 3 reveals that the AUC values for all classes are approximately equal to 1, underscoring the PSO-XGBoost model's excellent performance in distinguishing between classes and making accurate predictions for most URLs.

Next, we analyzed the model's performance across datasets of different sizes. Table 7 provides a summary of accuracy and Kappa results for varying dataset sizes.

As demonstrated in Table 7, there is a consistent upward trend in both accuracy and kappa values as the dataset size increases. This trend indicates that the PSO-XGBoost model exhibits improved performance as the dataset expands, suggesting that even higher accuracy levels can be achieved with larger datasets. To provide a comprehensive analysis of the model's performance across datasets of varying sizes, we conducted a comparison of precision and F-measure rates, with the results depicted in Fig. 4.

Fig. 4 illustrates the assessment of our proposed model's performance across datasets of varying sizes, focusing on precision and F-

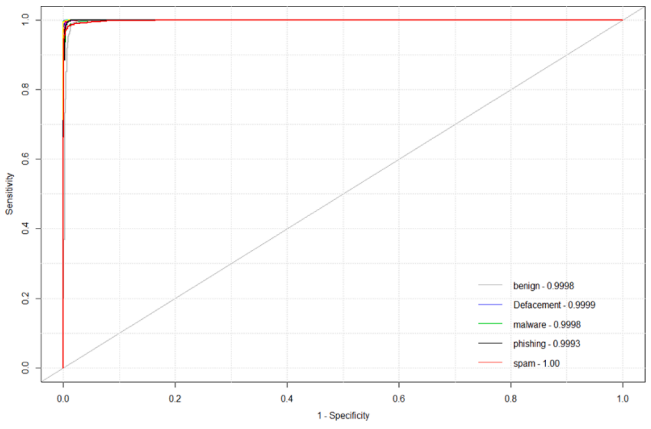


Fig. 3. ROC curve for different classes.

Table 7
Comparison of Accuracy and Kappa results on different size of datasets.

No. of Items	Accuracy	Kappa
7304	95.36	94.20
14,685	97.07	96.33
22,026	97.50	96.87
29,367	97.72	97.14
36,707	98.42	98.02

measure rates. The data depicted in this figure demonstrates a progressive improvement in precision, F-measure rates, and accuracy levels, as outlined in Table 7. As anticipated, the PSO-XGBoost model consistently outperforms other models, mainly when applied to datasets containing a large number of websites. The efficacy of this model is further enhanced by our feature selection technique, resulting in more accurate identification of diverse malicious attacks within a shorter processing time.

In this study, we employed an extensive set of input features. The suggested PSO-XGBoost model consistently exhibited exceptional performance in accurately detecting a wide range of harmful URLs. Consequently, a thorough investigation into the significance of the input features utilized in this study is warranted. In our subsequent analysis, we assessed the importance of these selected features, as summarized in the following figure. The ranking of features is based on their importance, estimated from the data used in building our proposed model.

Fig. 5 presented the individual contributions of attributes to the model's predictive performance. The figure highlights "NumberofDotsinURL," "UrlLen," and "ArgDomanRatio" as the three most crucial attributes within the dataset based on their significant contributions. Subsequently, we provide a comprehensive description of these attributes and elucidate their significance in the detection of malicious websites:

NumberofDotsinURL: Legitimate websites typically utilize up to two levels of domains in their URLs. Any website containing more than three dots within its domain can be considered suspicious and potentially categorized as a malicious site.

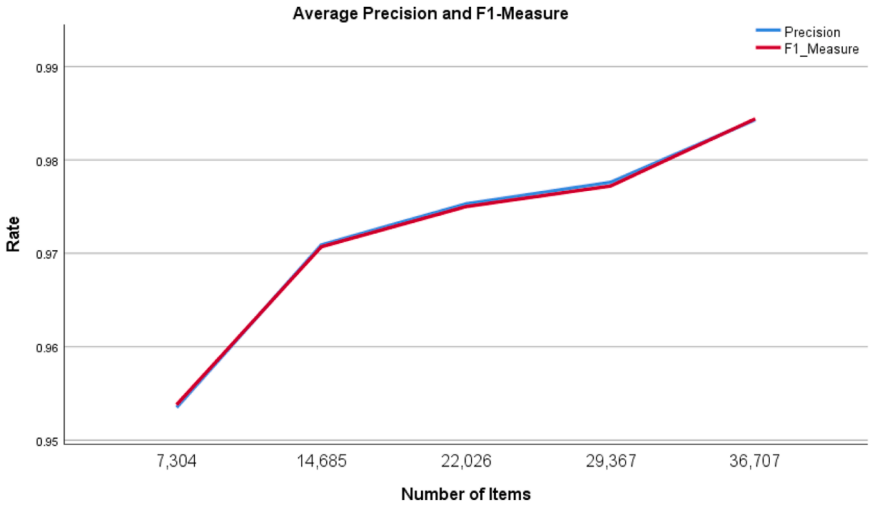


Fig. 4. The comparison of precision and F1-measure results on different size of datasets.

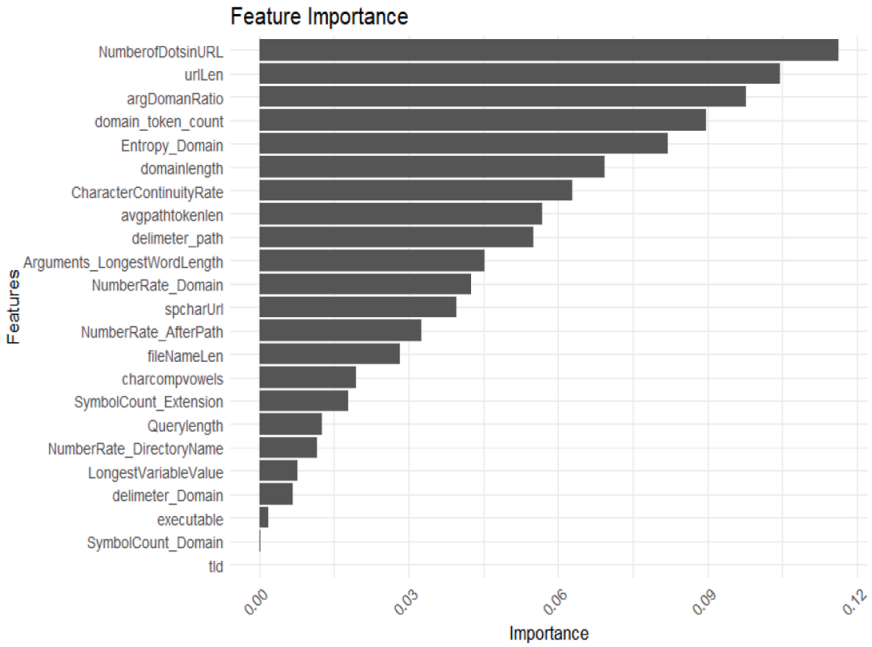


Fig. 5. Rank of Features by Importance in model.

UrlLen: This attribute quantifies the length of URLs. Generally, legitimate websites tend to have URLs of approximately 75 characters in length. Websites featuring unusually long URLs may raise suspicions and could be classified as malicious.

ArgDomanRatio: This attribute assesses the ratio of arguments to the domain. In many malicious web pages, this ratio is higher compared to legitimate websites.

As illustrated, these attributes play a vital role in the identification of diverse types of malicious websites and significantly contribute to the model's accuracy. Moreover, they serve as effective discriminators between legitimate and malicious websites, as most malicious entities aim to minimize costs and time when creating harmful web pages, leading to a notable absence of these variables in many malicious websites.

6. Conclusions

The internet is rife with malicious websites attempting to pilfer people's sensitive information through various forms of cyberattacks. These malicious sites pose critical challenges on the internet, hampering users from conducting their online activities safely. This study introduced a novel machine learning (ML) methodology designed to classify malicious websites accurately. Our proposed system consists of two primary phases. Initially, we employ a feature selection technique to identify the most relevant features, thereby streamlining complexity and processing time. Subsequently, we employ a Particle Swarm Optimization eXtreme Gradient Boosting (PSO-XGBoost) model to classify websites based on the selected features. To validate the efficacy of our proposed model, we conducted extensive experiments, comparing it against six benchmark machine learning algorithms using a real-world dataset comprising over 36,000 web pages from malicious websites. The results were compelling. In binary classification problems, our model successfully classified 98.42 % of web pages, and in multiclass problems, it achieved over 98 % accuracy for each class. These findings underscore the effectiveness and feasibility of our detection method, surpassing the performance of the benchmark algorithms in terms of accuracy and F1-measure.

One of the primary limitations of our study is the need for real-world data and the limited number of malicious web page categories due to the inadequacy of available datasets. Future research efforts should focus on collecting real-world data from various categories of malicious websites. Furthermore, we plan to enhance our model to address a broader spectrum of malicious activities. This study also highlights the potential of incorporating swarm intelligence applications into machine learning techniques for the successful detection of complex malicious websites. This methodology can also find applications in addressing other classification problems.

CRediT authorship contribution statement

Saeid Sheikhi: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Software, Project administration, Validation, Visualization, Writing – original draft, Writing – review & editing. **Panos Kostakos:** Project administration, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data is publicly available and is appropriately cited within the manuscript.

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